BCSC0004: OPERATING SYSTEMS

Objective: *This course aims to introducing the concept of computer organization. In particular, it focuses on basic hardware architectural issues that affect the nature and performance of software.*

Credits:03 **L-T-P-J:3-0-0-0**

Module No.	Content	Teaching Hours
I	**Introduction:** Operating System and its Classification - Batch, Interactive, Multiprogramming, Time sharing, Real Time System, Multiprocessor Systems, Multithreaded Systems, System Protection, System Calls, Reentrant Kernels, Operating System Structure- Layered structure, Monolithic and Microkernel Systems, Operating System Components, Operating System Functions and Services. **Processes:** Process Concept, Process States, Process State Transition Diagram, Process Control Block (PCB), Process Scheduling Concepts, Threads and their management. **CPU Scheduling:** Scheduling Concepts, Performance Criteria, Scheduling Algorithms, Multiprocessor Scheduling. **Process Synchronization:** Principle of Concurrency, Implementation of concurrency through fork/join and parbegin/parend, Inter Process Communication models and Schemes, Producer / Consumer Problem, Critical Section Problem, Dekker's solution, Peterson's solution, Semaphores, Synchronization Hardware. **Classical Problem in Concurrency:** Dining Philosopher Problem, Readers Writers Problem.	20
II	**Deadlock:** System model, Deadlock characterization, Prevention, Avoidance and detection, Recovery from deadlock, Combined Approach. **Memory Management:** Multiprogramming with fixed partitions, Multiprogramming with variable partitions, Paging, Segmentation, Paged segmentation. **Virtual memory concepts:** Demand paging, Performance of demand paging, Page replacement algorithms, Thrashing, Locality of reference. **I/O Management and Disk Scheduling:** I/O devices, I/O subsystems, I/O buffering, Disk storage and disk scheduling. **File System:** File concept, File organization and access mechanism, File directories, File allocation methods, Free space management.	20

Text Books:

- Silberschatz, Galvin and Gagne, "Operating Systems Concepts",9th Edition, Wiley, 2012.

Reference Books:

- SibsankarHalder and Alex a Aravind," Operating Systems", 6th Edition, Pearson Education, 2009.
- Harvey M Dietel, "An Introduction to Operating System", 2nd Edition, Pearson Education, 2002.
- D M Dhamdhere, "Operating Systems: A Concept Based Approach", 2nd Edition, 2006.
- M. J. Bach, "Design of the Unix Operating System", PHI, 1986.

Course Name: Operating Systems

Course Outcome

CO1- Understand the classification of operating system environment
CO2- Understand the basic of process management
CO3- Apply the concept of CPU process scheduling for the given scenarios.
CO4- Illustrate the process synchronization and concurrency process in operating system
CO5- Analyze the occurrence of deadlock in operating system.
CO6- Describe and analyze the memory management and its allocation policies.
CO7- Understand the concepts of disk scheduling

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End Term Examination, Odd Semester 2022-23
B. Tech. (CSE / EC with minor in CSE), II Year, III Semester
Operating Systems (BCSC0004)

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No	Detail of Question	Marks	CO	BL	KI.
1	State the Reader Write problem and discuss its solution using Semaphore.	4	4	U	C
2	What is convoy effect? Consider the following set of processes that need to be scheduled on a single CPU. All the times are given in milliseconds. Process Name / Arrival Time / Execution Time A / 0 / 6 B / 3 / 2 C / 5 / 4 D / 7 / 6 E / 10 / 3 Using the *shortest remaining time first* scheduling algorithm, calculate the average process turnaround time (in msec).	4	3	A	P
3	What is the main purpose of an operating system? Also, explain the advantages and applications of Multithreaded System?	4	1	U	C
4	Given 3 processes A, B and C, three resources x, y and z and following events, i) A requests x ii) A requests y iii) B requests y iv) B requests z v) C requests z vi) C requests x vii) C requests y Assume that requested resources should always be allocated to the request process if it is available. Draw the	4	5	An	P

	resource allocation graph for the sequences. And also mention whether it is a deadlock? If it is, how to recover the deadlock.				
5	What is seek time? Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive currently services a request at cylinder 143, and the previous request was at cylinder 125. the queue of pending request in FIFO order is: 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 Starting from the current position, what is the total distance (in cylinders) that the disk arm moves to satisfy all pending requests, for each of the following algorithms i) SSTF ii) C-SCAN.	4	7	A	P

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Explain Belady's Anomaly with an example. A system uses FIFO policy for page replacement. It has 4-page frames with no pages loaded to begin with. The system first accesses 100 distinct pages in some order and then accesses the same 100 pages but now in the reverse order. How many page faults will occur?	3	7	An	P
7	What is paging? Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 milliseconds to search the TLB and 80 milliseconds to access the physical memory. If the TLB hit ratio is 0.6, the effective memory access time (in milliseconds) is	3	6	U	C
8	In a virtual memory system, size of virtual address is 32-bit, size of physical address is 30-bit, page size is 16 Kbyte and size of each page table entry is 32-bit. The main memory is byte addressable. Which one of the following is the maximum number of bits	3	6	An	M

	that can be used for storing protection and other information in each page table entry?				
9	Differentiate between internal fragmentation and external fragmentation with example.	3	5	U	R
10	Explain the different methods for free space management with an example.	3	7	An	C

Section – C

Attempt All Questions

5 X 3 = 15 Marks

No	Detail of Question	Marks	CO	BL	KL
11	Discuss the segmentation memory management technique with an example? Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. Perform the allocation of processes using- 1. First Fit Algorithm 2. Best Fit Algorithm 3. Worst Fit Algorithm	5	6	A	P
12	Explain different file allocation methods with their advantages and disadvantages.	5	7	R	C
13	Differentiate between virtual address and physical address. Consider the virtual page reference string 1, 2, 3, 2, 4, 1, 3, 2, 4, 1 On a demand paged virtual memory system running on a computer system that main memory size of 3 pages frames which are initially empty. Find the number of page faults under the corresponding page replacements policy: LRU, FIFO and OPTIMAL.	5	6	An	M

Course Name: Operating Systems

Course Outcome

CO1- Understand the classification of operating system environment.
CO2- Understand the basic of process management.
CO3- Apply the concept of CPU process scheduling for the given scenarios.
CO4- Illustrate the process synchronization and concurrency process in operating system.
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End Term Examination, Even Semester 2021-22
B. Tech. (CSE), II Year, IV Semester
Operating Systems (BCSC0004)

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	State the Dining Philosopher problem. Discuss the solution to Dining Philosopher problem using semaphore.	4	4	U	C
2	Consider the following table of arrival time and burst time for three processes P_0, P_1 and P_2. Process / Arrival time / Burst time P_0 / 0 ms / 9 ms P_1 / 1 ms / 4 ms P_2 / 2 ms / 9 ms Scheduling is carried out only at arrival or completion of a process. Calculate the average waiting time for pre-emptive shortest job first scheduling algorithm for the three processes?	4	3	A	P
3	Discuss all the services provided by the Operating System to both the users and to the programs.	4	1	U	C
4	What are the necessary conditions for a deadlock to arise? Explain them.	4	5	R	F

5	Suppose the following disk request sequence (track numbers) for a disk with 100 tracks is given: 45, 20, 90, 10, 50, 60, 80, 25, 70. Assume that the initial position of the R/W head is on track 50. Calculate the additional distance that will be traversed by the R/W head when the Shortest Seek Time First (SSTF) algorithm is used compared to the SCAN (Elevator) algorithm (assuming that SCAN algorithm moves towards 100 when it starts execution).	4	7	A	P

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What is safe state in deadlock? A system shares 9 tape drives. The current allocation and maximum requirement of tape drives for three processes are shown below: Process / Current Allocation / Maximum Requirement P1 / 3 / 7 P2 / 1 / 6 P3 / 3 / 5 Determine whether the current state of the system is safe or not?	3	5	A	P
2	Discuss the advantages of Dynamic Partitioning over fixed partitioning memory management technique.	3	6	U	C
3	Consider a computer system with 40-bit virtual addressing and page size of sixteen kilobytes. If the computer system has a one-level page table per process and each page table entry requires 48 bits, then calculate the size of the per-process page table in megabytes.	3	6	An	M

4	What are the various attributes that are associated with an opened file? Explain the single level directory structure with its advantages and disadvantages.	3	7	R	F
5	Distinguish between Logical and Physical address. A paging scheme uses a Translation Lookaside buffer (TLB). The effective memory access takes 190 ns and a main memory access takes 50 ns. What is the TLB access time (in ns) if the TLB hit ratio is 90% and there is no page fault?	3	6	An	C

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider the following snapshot of the system: Find out the following by applying Banker's Algorithm: a) How many resources are there of type (A, B, C)? b) What are the contents of the Need matrix? c) Is the system in a safe state? Why?	5	5	A	P
2	What is Thrashing? Discuss the effects and techniques to handle the Thrashing.	5	6	R	C
3	Explain the Demand Paging. Consider a main memory with five-page frames and the following sequence of page references: 3, 8, 2, 3, 9, 1, 6, 3, 8, 9, 3, 6, 2, 1, 3. Calculate the page fault for the First-In-First Out (FIFO), Least Recently Used (LRU) and Optimal page replacement policies?	5	6	An	M

	Max			Allocation			Available		
	A	B	C	A	B	C	A	B	C
P0	0	0	1	0	0	1	1	5	2
P1	1	7	5	1	0	0			
P2	2	3	5	1	3	5			
P3	0	6	5	0	6	3			

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End Term Examination, Odd Semester 2021-22

BTech (CSE) - II Year / III Semester

Operating System (BCSC0004)

Time: 3 Hour **Maximum Marks: 50**

Section- A

Note: Attempt any FIVE Questions. **5 x 4 = 20 marks**

1. What is Semaphore? Explain in detail the Reader & Writer Problem of process synchronization with suitable code (both Reader and Writer Process).
2. Explain Pre-emptive and Non Pre-emptive CPU scheduling algorithm? Describe Multilevel feedback queue scheduling in brief!
3. Describe Inter Process Communication (IPC) and their method to implement IPC in the system?
4. How deadlock condition occurs? What are the necessary and sufficient condition for a system to be in deadlock? Explain in detail!
5. The Logical memory and physical memory is 32k and 512 words respectively. If Page size is of 64 words, formulate logical and physical memory space. How many pages logical and physical memory can accommodate? Write number of entries in the page table also.
6. Write the possible protocols to prevent "**Hold and Wait**" and "**Circular Wait**" condition for Deadlock Prevention? Clearly Explain!

Section- B

Note: Attempt ALL Questions. **5 x 3 = 15 marks**

1. What is external fragmentation? What are the probable solution for the removal of external fragmentation? Explain!
2. Considering a system with five processes P_0 through P_4 and three resources types A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Suppose at time t_0 following snapshot of the system has been taken:

Process	Maximum Need			Allocation		
	A	B	C	A	B	C
P0	7	5	3	0	1	0
P1	3	2	2	2	0	0
P2	9	0	2	3	0	2
P3	2	2	2	2	1	1
P4	4	3	3	0	0	2

a. Find the Need Matrix?
b. Find whether system is in Deadlock or not? Write the safe sate (if Possible)
c. What will happen if process P_1 requests one additional instance of resource type A and two instances of resource type C? Is additional request by process P1 granted?

3. A virtual memory system has an address space of 16K words, a memory space of 6K words, and page and block sizes of 2K words The following page reference changes occur during given time interval.

2 4 0 1 2 6 1 4 0 1 0 2 3 5 7

Determine the three pages that are resident in main memory after each page reference change if the replacement algorithm used is (a) FIFO; (b) LRU.

4. Given five memory partitions of 500 KB, 350 KB, 250 KB, 420 KB, and 450 KB (in order), how would the, best fit and worst-fit algorithms place processes of 325 KB, 150 KB, 400 KB, and 375 KB (in order)?
5. Write short notes on the following
   a. Demand Paging
   b. Thrashing

Section- C

Note: Attempt any THREE Questions. **3 x 5 = 15 marks**

1. What is file System? Explain various of directory structure with suitable example and diagram.

2. What is safe sequence? Write and explain Banker's Algorithm for Dead avoidance.
3. Draw the resource allocation graph and determine deadlock or deadlock free state.

$P = \{P_1, P_2, P_3\}$
$R = \{R_1, R_2, R_3, R_4\}$
$E = \{ P_1 \rightarrow R1, P2 \rightarrow R3, R1 \rightarrow P2, R2 \rightarrow P2, R2 \rightarrow P1, R3 \rightarrow P3, P3 \rightarrow R2\}$
There are ONE, TWO, ONE and THREE instances of R_1, R_2, R_3 and R_4 type resources

4. In a paged-segmented system, a virtual address consists of 32 bits of which 12 bits are displacement, 11 bits segment number and 9 bits are page numbers.
   Calculate the following:
   a. Page size
   b. Max segment size
   c. Max number of Pages
   d. Max number of segments

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End Term Examination, Odd Semester 2019-20
B. Tech. (CSE), II-Year, III-Semester
BCSC 0004 : Operating Systems

Time: 3 Hours **Max Marks: 50**

Note :
1. Assume suitable data.
2. Draw neat diagrams wherever necessary.

Section – A

Note : Attempt all question. **(7x5=35)**

I. Compare Monolithic, Layered, and Microkernel Architecture of Operating System.

II. Draw the Gantt chart and calculate the turnaround time and waiting time for the following processes using SJF Preemptive and Round Robin CPU scheduling algorithms (Consider time quantum = 2)

Process	Arrival Time	Burst Time
P1	0	6
P2	1	4
P3	4	8
P4	3	3

III. What is Deadlock? State and Explain necessary conditions for Deadlocks.

OR

Consider the following snapshot of the system:

Process	Allocation			Max			Available		
	R1	R2	R3	R1	R2	R3	R1	R2	R3
P1	0	1	0	7	5	3	3	3	2
P2	2	0	0	3	2	2			

P3	3 0 2	9 0 2	
P4	2 1 1	2 2 2	
P5	0 0 2	4 3 3	

a) What are the values of need Matrix

b) Is the system in safe state? If Yes, what is the safe sequence?

c) If a request from process P1 arrives for (1,0,2), can the request be granted immediately.

IV. Define Strong and Weak Semaphore. Describe the deadlock free solution for Dining Philosopher Problem using semaphores.

V. Describe Process Control Block with the help of neat diagram. Discuss what role does it play during context switching. Also, List the CPU scheduling criteria.

VI. For the following reference string ,determine the number of page faults that occur with 3 frames using LRU, and Optimal page replacement policy:
1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 3, 6.

VII. For the give sequence of disk request, calculate the total distance travelled by disk head in satisfying the entire request for C-SCAN, C- LOOK and SSTF Disk scheduling algorithms. Initial head position is 100 and total number of cylinders in the disk is 200.
27, 129, 110, 186, 147, 41, 10, 64, 120

Section – B

Note : Attempt all questions. **(3x2=6)**

I. Define the following terms:
a) Seek Time
b) Rotational Latency

II. Describe Indexed Sequential File Access Method.

III. How the free space management is done in Operating System.

Note : Attempt all questions. **(3x3=9)**

I. List and Explain I/O Buffering Techniques.

II. Compare and Describe First Fit, Best Fit and Next Fit memory allocation algorithms with the help of example.

III. Describe the following terms:

a) Principle of Locality

b) Thrashing

c) External Fragmentation.

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End Term Examination, Odd Semester 2018-19

Programme: B-Tech **Branch:** CSE **Year:** II

Subject: Operating Systems, BCSC 0004

Time: 3 Hours **Maximum Marks: 50**

Section-A

Attempt all Questions **7 x 5 = 35 Marks**

I. Discuss different types of operating systems with examples. Also comment on the validity of following statement "An operating system creates virtual devices when it is short of real devices".

II. What is the main advantage for an operating system designer of using a virtual machine architecture? What are the advantages for the user of using this architecture?

Or

II. Explain different types of access methods for files. Suppose, a file named "data" is frequently accessed by users in a system. The following alternatives are proposed to simplify access to data.

a. Set up links from every user's home directory to data.

b. Copy data into every user's home directory.

Compare the advantages and drawbacks of these approaches.

III. What are the benefits of multithreading programming? Explain different types of multithreading models.

IV. Round-robin schedulers normally maintain a list of all runnable processes, with each process occurring exactly once in the list. What would happen if a process occurred twice in the list? Can you think of any reason for allowing this? Most of the Round – robin schedulers uses a fixed time quantum. Give an argument for small and large time quantum.

V. Explain hardware based solution to the synchronization problem. Also show that, if both the wait () and signal() semaphore operations are not executed atomically, then mutual exclusion may be violated.

Or

V. Two concurrent processes share a data item sum, which is initialized to 0. However, they do not use mutual exclusion while accessing its value. Each process contains a loop that executes 50 times and contains the single statement sum =

sum+1. If no other operations are performed on sum, indicate the lower bound and upper bound on the value of sum when both processes terminate. Also, explain your reasoning.

VI. Five batch jobs. A through E, arrive at the CPU at almost the same time. They have estimated running times of 10, 6, 2, 4, and 8 minutes. Their (externally determined) priorities are 3, 5, 2, 1, and 4, respectively, with 5 being the highest priority. For priority and round- robin scheduling (time quantum = 2) algorithms, determine the average turnaround time, average response time, average waiting time.

VII. Why segmentation and paging sometimes combined in one section? Consider the following segment table:

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses for the following logical addresses?

a) 0,430
b) 1,10
c) 2,500
d) 3,400
e) 4,112

Section-B

Attempt All Questions **3 x 2 = 6 marks**

I. What kind of hardware support OS need to implement TLB? Is it possible to increase TLB size of a computer by upgrading or updating the OS? (2)

II. Consider a swapping system in which memory consists of the following hole sizes in memory order: 10 MB, 4 MB, 20 MB, 18 MB, 7 MB, 9 MB, 12 MB, and 15 MB. Which hole is taken for successive segment requests of

(a) 12 MB
(b) 10 MB
(c) 9 MB

for first fit? Now repeat the question for best fit, worst fit, and next fit. (2)

III. What is the cause of thrashing? How can you model thrashing based on working set of processes? (2)

Attempt All Questions **3 x 3 = 9 marks**

IV. A computer system has a 36-bit virtual address space with a page size of 8K, and 4 bytes per page table entry.

a) How many pages are in the virtual address space?
b) What is the maximum size of addressable physical memory in this system?
c) If the average process size is 8GB, then compute the average size of a page table for a two-level page table? (Use P1= 12 & P2= 11 in page field of Virtual Address) (3)

V. Disk requests come in to the disk driver for cylinders 10, 22, 20, 2, 40, 6, and 38, in that order. A seek takes 6 msec per cylinder. How much seek time is needed for

(a) First-come, first served.
(b) SSTF
(c) Elevator algorithm (initially moving upward).

In all cases, the arm is initially at cylinder 20. (3)

VI. Consider the following page reference string:

1, 2, 3, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6.

How many page faults will occur if FIFO, LRU and Optimal page replacement policy are used? Assume there are 4 page frames and initially all are empty. (3)

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End Term Examination, Even Semester 2017-18

Course : B.Tech.(CSE), Year : II, Semester : IV

Subject Code : **CSE4001** Subject Title : **Operating Systems**

Time : **02 ½ Hours** Max. Marks: **40**

Section- A

Note: Attempt All Questions. 1 x 16 = 16 marks

(I) What is the difference between Hard real time and Soft real time operating system?

(II) Enumerate three UNIX system calls used in Process Control.

(III) What are the advantages of the layered approach to the design of operating system?

(IV) What is the difference between preemptive and non-preemptive scheduling?

(V) The following are some events that occur after a device controller issues an interrupt while process L is under execution.

A. The processor pushes the process status of L onto the control stack.

B. The processor finishes the execution of the current instruction.

C. The processor executes the interrupt service routine.

D. The processor pops the process status of L from the control stack.

E. The processor loads the new PC value based on the interrupt.

What is the correct order in which the above events occur?

(VI) Differentiate between User-level thread and Kernel-level thread.

(VII) Enlist the parameters by which a multilevel feedback queue scheduler is defined.

(VIII) What is the role of processor affinity in context to multiple-processor scheduling?

(IX) What are the requirements that a solution to critical-section problem must satisfy?

(X) What are the two protocols by which we can ensure hold-and-wait condition never occurs in the system?

(XI) A system contains 6 units of resource, and n processes that use the resource. What is the maximum value of n for which the system will be deadlock free if the maximum requirement of each process is 3?

(XII) "Counting semaphores can be used to control access to a given resource consisting of a finite number of instances." Justify the statement.

(XIII) Differentiate between internal and external fragmentation.

(XIV) What is Belady's anamoly? Which algorithm suffers from it?

(XV) Consider a computer system with 48-bit virtual addressing and page size of 32 kilobytes. If the computer system has a one-level page table per process, then find out the no. of bits needed to represent page number in the page table.

(XVI) What is the purpose of ACLs in context to file system?

Section- B

Note: Attempt Any Four Questions. 3 x 4 = 12 marks

(I) When do page faults occur? Describe in detail the actions taken by the operating system when a page faults occur.

(II) Consider the reference string

1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6

How many page faults will occur for:

a. First in First out

b. LRU page replacement algorithm?

Assume three & four frames (initially empty) in each case.

(III) Consider a disk pack with a seek time of 4 milliseconds and rotational speed of 10000 rotations per minute (RPM). It has 600 sectors per track and each sector can store 512 bytes of data. Consider a file stored in the disk. The file contains 2000 sectors.

Assume that every sector access necessitates a seek, and the average rotational latency for accessing each sector is half of the time for one complete rotation. What is the total time (in milliseconds) needed to read the entire file?

(IV) What is a Directory? Explain any two ways to implement the directory.

(V) Write short notes on :

a. I/O Buffering

b. Sequential file

c. Indexed file

Section- C

Note: Attempt Any Three Questions. 4 x 3 = 12 marks

(I) Discuss the paged segmentation scheme of memory management and explain how logical address is translated to physical address in such a scheme.

(II) What is the cause of thrashing? How does system detect thrashing? Once it detects thrashing, what can system do to eliminate this problem?

(III) A computer uses 46–bit virtual address, 32–bit physical address, and a three–level paged page table organization. The page table base register stores the base address of the first–level table (T1), which occupies exactly one page. Each entry of T1 stores the base address of a page of the second–level table (T2). Each entry of T2 stores the base address of a page of the third–level table (T3). Each entry of T3 stores a page table entry (PTE). The PTE is 32 bits in size. The processor used in the computer has a 1 MB 16-way set associative virtually indexed physically tagged cache. The cache block size is 64 bytes. What is the size of a page in KB in this computer?

(IV) Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, … , 199), and 256 sectors per track (numbered as 0, 1, … 255). The following 6 disk requests of

the form [sector number, cylinder number, platter number] are received by the disk controller at the same time:

[120, 72, 2], [180, 134, 1], [60, 20, 0], [212, 86, 3], [56, 116, 2], [118, 16, 1]

Currently head is positioned at sector number 100 of cylinder 80, and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 40 milliwatts and for reversing the direction of the head movement once is 20 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible. What is the total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm?

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End Term Examination, Even Semester 2016-17
B. Tech., II-Year, IV-Semester
CSE 4001 : Operating System

Time: 2½ Hours Max. Marks: 40

Section – A

Attempt All Questions. **(1x16=16)**

I. What is dispatcher?

II. What are real time operating systems?

III. What is the main advantage of multiprogramming?

IV. Define Preemptive Kernel?

V. Mention any two properties of CPU scheduling algorithms.

VI. What is the advantage of Microkernel approach to system design?

VII. What do you understand by race-condition?

VIII. What is the use of semaphore?

IX. What are the two basic models of Inter Process Communication?

X. When will you say that the system is in safe state?

XI. What are the approaches used to handle Critical Section?

XII. What are monitors?

XIII. What is a virtual address?

XIV. Mention any two techniques used to solve the external fragmentation?

XV. Why the page sizes always powers of 2?

XVI. What is meant by thrashing?

Section – B

Attempt Any Four Questions. **(3x4=12)**

I. Consider a logical address space of eight pages of 1024 words each, mapped onto a physical memory of 32 frames.
   a. How many bits are there in the logical address?
   b. How many bits are there in the physical address?

II. Explain the various aspects of disk management briefly?

III. Given five memory partitions of 100kB, 500kB, 200kB, 300kB, and 600kB (in order), how would each of the first-fit, best-fit and worst-fit algorithms place processes of 212kB, 417kB, 112kB, and 426kB (in order)? Which algorithm makes the most efficient use of the memory?

IV. Discuss in detail about demand paging.

V. Why are segment and paging sometimes combined into one schema? Explain.

Section – C

Attempt Any Three Questions. **4 x 3 = 12 marks**

I. On a disk with 1000 cylinders, numbers 0 to 999, compute the number of tracks the disk arm must move to satisfy all the requests in the disk queue. Assume the last request serviced was at track 345 and head is moving to track 0. The queue in FIFO order contains requests for the following tracks: 123, 874,692,

475, 105, 376. Perform the computation for the following disk scheduling algorithms.

a.) FIFO b). SSTF c.) SCAN d.) LOOK

II. What is page fault? Consider the reference string 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5. There are 3 frames allotted in the memory at a time. Apply FIFO, LRU, Optimal page replacement algorithms and find the number of page faults.

III. Explain the various issues of storage allocation in a file system with that of main memory allocation.

IV. Explain the following:

a. Contiguous Allocation and Linked Allocation

b. Common schemes for defining the Logical structure of a directory.

END Term Examination, 2015-16

Course: B.Tech. (CSE), Year: II, Semester: IV

Operating System (CSE 4001)

Uni. Roll No:.....

Time: 2.5 Hrs. **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16)**

I. List the four steps that are necessary to run a program on a completely dedicated machine.

II. Write any two differences between trap and an interrupt.

III. What are the main advantages of the microkernel approach to system design?

IV. Differentiate Multiprogramming and Multitasking. (Any two)

V. What are the two conditions under which a parent process may terminate the execution of one of its children process?

VI. Explain the significance of PCB when CPU switches from one user to another.

VII. What are the fundamental requirements to be met by concurrent processes in order to cooperate correctly using the shared data?

VIII. What is race condition?

IX. Explain binary semaphore.

X. Define livelock.

XI. What is RAG?

XII. Define the need of Mutual Exclusion.

XIII. What do you mean by load time binding?

XIV. How can we reduce the memory overhead of a paged system?

XV. Why is page size always a power of 2?

XVI. What is thrashing?

Section B

Note: Attempt any Four Questions **(3x4=12)**

I. Under what circumstances do page fault occurs? Describe the actions taken by the operating system when a page fault occurs.

II. What hardware support are needed to implement virtual memory? Given a 4GB of virtual space and typical page size of 4KB. How many virtual pages would this imply? If each page table entry is 5 bytes, how much space is required for the whole page table?

III. Describe various file allocation strategies. Also discuss the criteria that have to decide which strategy is to be adopted for a particular file.

IV. Compare Paging and Segmentation. For a paged system, TLB hit ratio is 0.9. let the RAM access time, t be 20ns and the TLB access time, T be 100ns. Find out-

a) Effective memory access with TLB

b) Effective memory access without TLB

V. Given memory partitions of 10K, 50K, 20K, 30K, and 60K(in order), how would each of the algorithm First-fit, Best-fit and Worst-fit place processes of 21K, 41K, 11K

and 42K (in order)? Which algorithm makes the best use of memory?

Section C

Note: Attempt any Three Questions **(4x3=12)**

I. Explain Linked and Indexed allocation method of disk. Also discuss their merits and demerits. Suppose that the disk rotates at 3600 rpm.

a) What is the average rotational latency of this disk drive?

b) What seek distance can be covered in the time that you found for part (a)?

II. Suppose that a disk drive has 500 cylinders, numbered from 0 to 499. The drive is currently serving a request at cylinder 154, and the previous request was at cylinder 102. The queue of pending requests in FIFO order is 80, 147, 291, 177, 94, 409, 102, 350, and 130 starting from the current head position. What is the total head distance (in cylinders) that the disk arm moves for the following disk scheduling algorithms:

a. SSTF

b. LOOK

III. What do you understand by Belady's Anomaly? Which popular page replacement algorithm suffers from the Belady's Anomaly? Consider the following page reference string-

1, 2, 3, 1, 4, 5, 6, 2, 1, 3, 2, 7, 6, 3, 4, 1, 2, 6

How many page faults would occur for the following replacement algorithms assuming 3 frames?

a) LRU

b) Optimal

IV. Explain the difference between Internal & External fragmentation? In a paged segmented system, a virtual address consists of 32 bits, of which 12 bits are used for displacement, 11 bits are a segment number and 9 bits are a page number. Calculate-

a) Page Size

b) Maximum No. of Pages

c) Maximum Segmented Size

d) Maximum No. of Segments

Printed Pages: 03 University Roll No. .....

End Term Examination, 2014-15

B. Tech. II-Year, IV-Semester

CSE 221 : Operating Systems

Time: 2½ Hours **Max Marks:** 40

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section – A

Note: Attempt All Questions **(1x16=16)**

I. Why do you think that idleness in CPU occurs?

II. What are the main advantages of microkernel approach to system design?

III. What are the essential and desirable properties of an OS?

IV. Define PCB?

V. What are the different scheduling criteria?

VI. Differentiate between pre-emptive and non pre-emptive scheduling.

VII. Describe IPC?

VIII. Explain the term bounded waiting.

IX. Consider two concurrently running processes: P1 with a statement S1 and P2 with a statement S2. Suppose we require that S2 be executed only after S1 has completed. How can you implement this scheme readily by letting P1 and P2 shares a common semaphore synch, initialized to 0.

X. Explain spinlock semaphore.

XI. Explain Safe State.

XII. How a deadlock can be recovered.

XIII. Define Belady's Anomaly.

XIV. What is TLB?

XV. Define Latency and seek time with respect to disk I/O.

XVI. Define Overlays.

Section – B

Note: Attempt any Four Questions **(3x4=12)**

I. Consider the following segment table

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses of the following logical addresses?

i. 0, 430

ii. 1, 10

iii. 2, 500

iv. 3, 400

v. 4, 112

II. What is thrashing? What is the reason for it? How does Working-Set Model solve the problem of thrashing?

III. Describe the hardware that is required to support paging with an acceptable amount of overhead. Why are segmentation and paging sometimes combined into one scheme?

IV. Explain various types of directory structures with their merits and demerits.

V. Given a memory partitions of 100K, 500K, 200K, 300K, 600K (in order), how would each of the First-fit, Best-fit and Worst-fit algorithm place processes of 212K, 417K, 112K AND 426K (in order)? Which algorithm makes the efficient use of memory?

Section – C

Note: Attempt any Three Questions **(4x3=12)**

I. How many page faults will occur for the following reference string with 3 page frames using LRU and Optimal Page Replacement Algorithms?

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

II. Explain the concept of multiprogramming with fixed partition along with the two versions and multiprogramming with variable partition. Explain the two methods by which the memory waste in multiprogramming with variable partition can be handled.

III. What are the basic operations of a file system? Discuss linked allocation and index allocation scheme for a file allocation and compare index allocation scheme with contiguous allocation scheme.

IV. Suppose that a disk drive has 5000 cylinders numbered 0 to 4999. The drive is currently serving a request at cylinder 143 and the previous request was at cylinder 125, the queue of pending requests in FIFO order is

86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

Starting from current head position what is the total distance that the disk arm moves to satisfy all the pending requests for SCAN and C-LOOK disk scheduling algorithms.

Printed Pages-3 **University Roll No.....**

B. Tech. II Year IV Semester, End Term Examination, 2013-14

CSE 221: Operating Systems

Time: 02 ½ Hours **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16 Marks)**

I. Define Operating Systems.

II. Write the fundamental difference between asymmetric and symmetric multiprocessing.

III. List the four major categories of the benefits of multithreaded programming.

IV. Differentiate between I/O-bound and CPU-bound processes.

V. What is a bootstrap program, and where is it stored?

VI. What do you mean by degree of multiprogramming?

VII. What effect does the size of the time quantum have on the performance of an RR algorithm?

VIII. What role does the dispatcher play in CPU scheduling?

IX. Differentiate between preemptive and non preemptive scheduling.

X. What is the Critical Section Problem?

XI. What are the three general ways that a deadlock can be handled?

XII. What is the difference between deadlock prevention and deadlock avoidance?

XIII. What do you mean by virtual memory?

XIV. What is thrashing?

XV. Define Belady's Anomaly.

XVI. Define Latency and seek time with respect to disk I/O.

Section B

Note: Attempt any Four Questions **(3x4=12 Marks)**

I. Given a memory partitions of 100K, 500K, 200K, 300K, 600K (in order), how would each of the First-fit, Best-fit and Worst-fit algorithm place processes of 212K, 417K, 112K AND 426K (in order)? Which algorithm makes the efficient use of memory?

II. In a paged segmented system, a virtual address consists of 32 bits of which 12 bits are used for displacement, 11 bits are a segment number and 9 bits are a page number. Calculate. **a.** Page size **b.** Maximum number of pages **c.** Maximum segment size **d.** Maximum number of segments.

III. In a demand page memory 200 ns are required to satisfy memory request if the page is in memory. If a page is not in memory the request takes 7ms, then calculate the effective access time if the page fault rate is 5%.

IV. Differentiate between internal and external fragmentation. Which one occurs in paging? Which one occurs in segmentation?

V. Explain various types of directory structures with their merits and demerits.

Section C

Note: Attempt any Three Questions **(4x3=12 Marks)**

I. What is paging? Why is it used? What is the hardware support required for paging? Describe this with the help of a diagram.

II. How many page faults will occur for the following reference string with 4 page frames using LRU and Optimal Page Replacement Algorithms.

1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 9, 7, 8, 9, 5, 4, 5, 4, 2

III. Suppose that a disk drive has 5000 cylinders numbered 0 to 4999. The drive is currently serving a request at cylinder 143 and the previous request was at cylinder 125, the queue of pending requests in FIFO order is

86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

Starting from current head position what is the total distance that the disk arm moves to satisfy all the pending requests for SSTF and LOOK disk scheduling algorithms.

IV. Differentiate between –

- Programmed I/O and Interrupt driven I/O.
- Absolute path name and Relative path name

SET-D

GLA University, Mathura

Course: -B.Tech. II-Year, End-Term (Even Sem.) Examination, 2012-13

Subject: - OPERATING SYSTEM (CSE-221)

Time:-2.30Hour **M.M:-40**

Notes:-

1. Answer all questions from Group A (which is) compulsory, Any Four from Group B and Any Four from Group C.
2. All questions of the particular group should be answered collectively at one place. All parts of a question should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

Group-A (All questions are compulsory) **16X0.5=08**

I. System calls are usually invoked by using

- An indirect jump
- A Software interrupt
- Polling
- A privileged instruction

II. If the length of time slice is made more than the burst time of the process with highest burst time then round robin scheduling behaves similar to:

- Priority scheduling
- FCFS scheduling
- SJF scheduling
- None of these

III. Locality of reference implies that the page reference being made by a process

- Will always be to the page used in the previous page reference
- Is likey to be to one of the pages used in the last few page references
- Will always be to one of the pages existing in memory
- Will always lead to page fault

IV. Dirty bit of a page in a page table

- Allows only read on a page
- Helps avoid unnecessary writes on a paging device
- Helps maintian LRU information
- Used for protection

V. A counting semaphore was initialized to 10. Then 6P (wait) operations and 4V (signal) operations were performed on semaphore. What is the resulting value of semaphore?

- 2
- 12
- 8
- 0

VI. A computer has 6 tape drives, with 'n' processes competing for them. Each process may need two drives. What is the maximum value of 'n' for the system to be deadlock free?

- 4
- 6
- 5
- 3

VII. Which of the following scheduling algorithm is non-preemptive?

- Round-Robin
- FCFS
- Multilevel queue scheduling
- Multilevel queue with feedback

VIII. Which of the following need not necessarily be saved on a context saved between processes?

- General purpose registers
- Program Counter
- Translation Look-Aside Buffer
- All of the above

IX. Suppose an OS uses variable length partitions for memory management. At some particular time a running process occupies a partition between physical address 1000 and 20000. What is the physical address for virtual address 3000?

- 23000
- 25000
- 13000
- 33000

X. Which of the following file allocation technique suffers from external fragmentation?

- Contiguous allocation
- Indexed allocation
- Linked allocation
- Both Contiguous and Indexed allocation

XI. A virtual memory system has 32-bit virtual and physical address space with page size of 2 KB. How many bits will be needed for index and off-set of page?

- 21, 11
- 32, 32
- 32, 8
- 11, 21

XII. What information does the presence of a cycle in resource allocation graph give?

- Deadlock will always occur
- Deadlock will never occur
- Deadlock may or may not occur
- All of the above

XIII. The correct matching for the following pairs:

(A) Disk Scheduling
(B) Batch Processing
(C) Time Sharing
(D) Interrupt processing

(1) Round Robin
(2) SCAN
(3) LIFO
(4) FIFO

- A-4, B3, C-2 and D-1
- A-2, B-4, C-1 and D-3
- A-3, B-4, C-2, and D-1
- A-2, B-1, C-4 and D-3

XIV. Listed below are some OS abstractions (in left column) and the hardware components (in right column). Match the correct pairs:

(P) Thread
(Q) Virtual Address Space
(R) File System
(S) Signal

(1) Interrupt
(2) Memory
(3) CPU
(4) Disk

- P-3, Q-2, R-4, S-1
- P-1, Q-2, R-3, S-4
- P-2, Q-4, R-3, S-1
- P-4, Q-1, R-2, S-3

XV. Using large block size in a fixed block size file system leads to:

- Poorer disk throughput and poorer disk space utilization
- Better disk throughput and better disk space utilization
- Poorer disk throughput but better disk space utilization
- Better disk throughput but poor disk space utilization

XVI. In which of the following page replacement policies does Belady's anomaly occur?

- LFU
- NRU
- FIFO
- LRU